STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents core microbiological culture media preparation, aseptic isolation, subculture purification, and stock preservation procedures used in HOPE Lab research and is PROVIDED ONLY FOR THE HOPE LAB MEMBERS for training and internal reference. External users should independently assess method suitability, verify current standards and chemical safety for reagents, and obtain required biosafety authorization and other essential approvals. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

Microbial Culture Media Preparation, Isolation, Purification, and Stock Preservation

PRINCIPLE

Foundational microbiology depends on the controlled preparation of sterile nutrient media, the aseptic recovery of viable microorganisms from complex matrices, the physical separation of single clonal populations through iterative subculturing, and the long-term preservation of verified pure cultures without genetic or phenotypic drift (Hossain et al., 2020; Hossain et al., 2024). Culture media provide defined carbon, nitrogen, mineral, and buffer systems—ranging from general-purpose enrichment media (e.g., Nutrient Agar, Tryptic Soy Broth) to selective and differential formulations (e.g., MRS for lactic acid bacteria, MacConkey for enteric Gram-negative bacilli).

Because only a fraction of environmental or host-associated microorganisms are readily culturable under any single combination of nutrients, temperature, pH, and redox potential, culture recovery inherently selects for organisms adapted to the chosen medium and incubation parameters. Strict adherence to sterilization validation, aseptic technique, colony purity verification, and dual-vial cryopreservation ensures experimental reproducibility and prevents cross-contamination.

Learning Outcomes

- Select and prepare general-purpose, selective, and differential culture media with proper pH adjustment and sterilization verification.
- Execute serial decimal dilutions and spread/pour/streak plating techniques to obtain well-isolated single colonies.
- Purify candidate microbial isolates via sequential single-colony subculturing and confirm purity via macroscopic and microscopic checks.
- Prepare working slants/plates and long-term glycerol cryopreserved stocks (-80°C) with complete strain provenance metadata.
- Identify and prevent common causes of medium caramelization, incomplete sterilization, swarming overgrowth, and stock viability loss.

OBJECTIVE & SCOPE

Objective: To standardize culture media preparation, primary sample processing, single-colony isolation, purity verification, and short- and long-term preservation of bacterial and yeast isolates in HOPE Lab (Paul et al., 2024; Hossain et al., 2024).

Scope: Applicable to aerobic and facultatively anaerobic bacteria (including lactic acid bacteria, Bacillus spp., and standard indicator strains) and non-fastidious yeasts handled under BSL-1 or BSL-2 containment. Strict anaerobes, obligate intracellular pathogens, and Risk Group 3 agents require specialized containment and dedicated SOPs.

CORE CONCEPTS & MEDIA CLASSIFICATION

- **Media Functional Categories:** • General-Purpose / Basal Media (e.g., Nutrient Agar/Broth, TSA/TSB): Support broad non-fastidious heterotrophic growth.
  • Selective Media (e.g., MRS with low pH/acetate, MacConkey, Mannitol Salt Agar): Contain inhibitory agents (bile salts, high NaCl, organic acids, or antibiotics) that suppress non-target flora.
  • Differential Media (e.g., Blood Agar, EMB, MacConkey): Incorporate indicators (pH dyes, chromogens, red blood cells) to distinguish metabolic traits such as lactose fermentation or hemolysis.
- **Serial Dilution & Colony Counting Range:** Ten-fold serial dilutions (10^{-1} to 10^{-8}) in sterile physiological saline (0.85% NaCl) or PBS reduce dense sample populations to countable levels (typically 30–300 CFU per standard 90 mm plate) while minimizing colony overlap.
- **Clonal Purification Principle:** A single macroscopic colony on a primary isolation plate may harbor co-localized satellite cells. At least two to three consecutive streak-plate passages from a single well-separated colony are mandatory before designating an isolate as axenic (pure).
- **Cryopreservation & Glycerol Protection:** Storage at -80°C in 15–20% (v/v) sterile glycerol prevents intracellular ice-crystal shearing and osmotic dehydration, preserving genetic and phenotypic stability over multi-year periods.

MATERIALS & EQUIPMENT

Reagents & Consumables

- Dehydrated culture media (Nutrient Agar/Broth, TSB/TSA, MRS Agar/Broth, MacConkey, etc.)
- Sterile diluents: Physiological saline (0.85% w/v NaCl) or PBS (pH 7.2–7.4)
- Molecular/analytical-grade glycerol (autoclaved 40–50% v/v stock for final 15–20% cryovials)
- Gram-staining reagents & 3% hydrogen peroxide (catalase check)
- Sterile 90 mm Petri dishes, 1.5–2.0 mL screw-cap cryovials, spreaders, & inoculation loops

Equipment & Instrumentation

- Calibrated Steam Autoclave (121°C, 15 psi) with biological/chemical sterilization indicators
- Class II Biosafety Cabinet (BSL-2) & Laminar Airflow Workstation
- Calibrated Analytical Balance & pH Meter (with temperature compensation)
- Temperature-controlled Water Bath (48–50°C for tempering molten agar) & Incubators (30°C, 37°C)
- -80°C Ultra-Low Temperature (ULT) Freezer, 4°C Refrigerator, & Compound Light Microscope

STEP-BY-STEP PROCEDURE

Step 1: Media Formulation, pH Adjustment & Sterilization

Weigh dehydrated medium components accurately according to manufacturer specifications and dissolve completely in distilled/deionized water with gentle heating and stirring (boil agar media briefly until completely clear before dispensing). Measure and adjust pH at 25°C using 1 M HCl or 1 M NaOH. Dispense into borosilicate

bottles (≤75% fill volume, caps loosened) and autoclave at 121°C (15 psi) for 15 minutes. Cool molten agar to 48–50°C in a water bath before pouring ~20–25 mL per sterile Petri dish inside a biosafety cabinet. Include an uninoculated sterility control plate from every batch.

Step 2: Sample Homogenization & Serial Decimal Dilution

Aseptically weigh or pipette a representative sample (e.g., 1.0 g or 1.0 mL) into 9.0 mL of sterile 0.85% saline or PBS (10^{-1} dilution) and homogenize thoroughly by vortexing or stomaching. Transfer 1.0 mL into 9.0 mL of fresh sterile diluent (10^{-2}) using a fresh sterile pipette tip, vortexing between each step, and continue through 10^{-6}–10^{-8} as appropriate for expected microbial load.

Step 3: Primary Plating (Spread-Plate or Pour-Plate)

For spread-plating, dispense 0.1 mL (100 µL) of selected dilutions onto pre-dried agar plates and distribute evenly with a sterile L-shaped spreader until absorbed. Allow plates to dry upright for 10–15 minutes, then invert and incubate at the target temperature (e.g., 30°C or 37°C) for 24–72 hours under appropriate atmospheric conditions (aerobic or microaerophilic/candle-jar for LAB).

Step 4: Colony Selection & Iterative Streak-Plate Purification

Inspect primary plates and record distinct colony morphologies (size, margin, elevation, opacity, pigmentation). Pick a single well-isolated colony using a sterile loop and perform a four-quadrant streak onto fresh selective and non-selective agar plates. Repeat single-colony subculturing for at least 2–3 consecutive passages until all colonies on a plate exhibit uniform morphology.

Step 5: Purity Confirmation & Preliminary Phenotypic Check

Confirm axenic purity before preservation by performing Gram staining and light microscopy (1000× oil immersion) to verify uniform cellular morphology and staining reaction, alongside a rapid catalase/oxidase check where applicable. Discard or re-streak any culture showing mixed cell morphologies.

Step 6: Working Stock & Cryopreserved (-80°C) Master Stock Preparation

Grow a single verified pure colony in appropriate broth to late-exponential/early-stationary phase. Mix 0.5–0.8 mL of dense culture with an equal volume of sterile 30–40% (v/v) glycerol in labeled screw-cap cryovials (final concentration 15–20% v/v glycerol). Gently invert to mix, equilibrate for 15–20 minutes at 4°C, and store duplicate Master and Working cryovials at -80°C. Log strain ID, source, medium, date, and passage number in the HOPE Lab Strain Registry.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Inadequate Control	Required Control Standard
Autoclave Cycle & Time	Over-autoclaving caramelizes sugars (toxic furfurals) & degrades agar	Sterilize at 121°C for 15 min; temper at 48–50°C before pouring
Medium pH Accuracy	Wrong pH alters selectivity (e.g., MRS) and inhibits target recovery	Calibrate pH meter daily; verify pH at 25°C prior to autoclaving
Plate Surface Moisture	Wet agar surface causes swarming/confluent bacterial growth	Dry poured plates laminar-flow/inverted before spread-plating
Pipette Tip Changing	Reusing tips across dilutions carries over high cell counts	Use a fresh sterile tip for every serial transfer & vortex well
Subculture Passages	Single-passage picks frequently retain hidden co-contaminants	Perform ≥2–3 consecutive quadrant streaks + Gram-stain check

Cryostock Freeze–Thaw	Repeated thawing of master vials kills cells & selects variants	Maintain separate Master & Working vials; scrape ice without thawing

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Incubate at least one uninoculated sterility control plate/tube from every autoclaved media batch (24–48 h at 37°C) before use. Verify selective and differential media performance using known positive and negative reference strains.

Observation	Possible Explanation	Corrective Action
Growth on uninoculated plate	Incomplete sterilization, contaminated supplement, or aseptic breach	Discard entire media batch; audit autoclave tape/biological indicator
Soft or non-solidifying agar	Low pH hydrolysis during autoclaving or incomplete agar dissolution	Boil agar fully before autoclaving; add acidulants after cooling
Confluent lawn / no single colonies	Dilution series insufficient or wet agar surface	Extend serial dilutions (10^{-6}–10^{-8}) and pre-dry agar plates
Mixed cell types on Gram stain	Co-isolated satellite contaminant on primary pick	Re-streak onto non-selective agar for 2–3 passes; re-verify purity
Failure to revive from -80°C stock	Improper glycerol %, slow freezing, or repeated vial thawing	Plate larger inoculum (100 µL) into enrichment broth; use fresh working vial

BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety & Containment:** Treat all unknown environmental, food, and clinical isolates as potential Risk Group 2 agents until identified. Perform all open manipulations of liquid cultures and aerosol-generating steps inside a certified Class II Biosafety Cabinet. Autoclave all used plates, tubes, and pipette tips at 121°C for 30 minutes prior to disposal.
- **Strain Provenance & Traceability:** Never label plates or cryovials on lids alone—always label tube walls and Petri dish bases with permanent solvent-resistant markers including Strain ID, Medium, Date, and Operator Initials.

REFERENCES & RELATED DOCUMENTS

Hossain TJ, Chowdhury SI, Mozumder HA, Chowdhury MNA, Ali F, et al. (2020). Hydrolytic exoenzymes produced by bacteria isolated and identified from the gastrointestinal tract of Bombay duck. Frontiers in Microbiology, 11, 1025. doi:10.3389/fmicb.2020.01025

Paul S, Hossain TJ, Ali F, Hossain ME, Chowdhury T, Faisal IK, Ferdouse J. (2024). Assessment of the in-vitro probiotic efficacy and safety of Pediococcus pentosaceus L1 and Streptococcus thermophilus L3 isolated from Laban, a popular fermented milk product. Archives of Microbiology, 206, 82. doi:10.1007/s00203-023-03812-5